

Jonathan Cruz

SENIOR BACKEND & PAYMENTS ENGINEER

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PROFILE

"Senior software engineer with over 10 years of experience building fintech and payments infrastructure. Deep expertise in transactional Postgres, double-entry ledgers, KYC/EDD compliance, and integrations with banking and identity providers (Plaid, Treasury Prime). Strong TypeScript across the stack with NestJS, Prisma, and event-driven architectures designed for fault tolerance, idempotency, and auditability."

PROFESSIONAL EXPERIENCE SELECTED ROLES & CONTRIBUTIONS

FINTECH FOR VC FUND ADMINISTRATION Sydecar FULL-STACK SOFTWARE ENGINEER	Double-Entry Ledger POSTGRES Contributed to a proprietary double-entry ledger engine (Fragment) with append-only immutable entries, deterministic idempotency keys (), derived balances (never stored as source of truth), and minor-unit integer accounting with banker's rounding. Posting flow is event-driven and fully reconstructable from source events. POSTGRESLEDGERDOUBLE-ENTRYDECIMAL Plaid + Treasury Prime BANKING Worked on Plaid Link integration (createLinkToken → exchangePublicKey → access_token) with processor tokens powering ACH transfers via Treasury Prime. Built a mock banking engine for the local sandbox with idempotency tracking by application/account IDs. PLAIDTREASURY PRIMEACHBANKING KYC/EDD COMPLIANCE Designed and implemented Enhanced Due Diligence including LP funding blocks, UBO verification, FinCEN integration, CTA compliance, and automatic KYC revalidation for active investors. Built race-resistant verify-identity flow with re-check inside a to handle "email-free-at-blur → taken-at-submit" edge cases (409 EMAIL_NOW_EXISTS). COMPLIANCEKYCUBOFINCEN Webhook Idempotency CONCURRENCY Implemented webhook dedup via (unique) with PROCESSING/PROCESSED/REJECTED state machine, and an API-level idempotency interceptor with 48h TTL scoped by . IDEMPOTENCYWEBHOOKSCONCURRENCY Fault Tolerance BULLMQ OUTBOX Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base in ledger; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs are idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR counters with deterministic predecessor lookup (). BULLMQOUTBOXREDISFAULT-TOLERANCE Billing V2 ARCHITECTURE Designed (ADR) and implemented fund billing V2 with quarterly fee scheduler, billing model versioning, and administrative fee structure management. TS-RESTZODNESTJS QuickBooks INTEGRATION Built bidirectional invoice sync engine with mock service for local dev, state validation, auto-sync on approval, and full E2E test suite with Playwright and Cucumber BDD. QUICKBOOKS APIPLAYWRIGHTCUCUMBER ts-rest TYPE-SAFE CONTRACTS Migrated legacy controllers (auth, onboarding, admin) to type-safe ts-rest contracts with Zod schemas, improving type safety end-to-end. TYPESCRIPTZODTS-REST
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B2B PAYMENTS & WHOLESALE Zax SR SOFTWARE ENGINEER JAN 2019 — MAY 2023 SÃO PAULO	B2B Payments MOBILE Built from scratch a wholesale ordering app with integrated payment flows and credit-limit validation for business partners, using React Native, TypeScript, and GraphQL. REACT NATIVETYPESCRIPTGRAPHQL Serverless AWS Deployed full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) with serverless web apps using React and Next.js. AWSNEXT.JSSERVERLESS CI/CD AUTOMATION Configured end-to-end delivery pipeline with GitHub Actions, Fastlane, and Code Climate for quality control. Docker for local dev. GITHUB ACTIONSFASTLANEDOCKER
CRYPTO CONSUMER FINTECH Netlolo SR SOFTWARE ENGINEER SEP 2018 — JAN 2019 SÃO PAULO	Crypto Wallet WEB3 Built a mobile + web app bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum for self-custodial wallets and on-chain transactions. REACT NATIVETYPESCRIPTWEB3ETHEREUM Chat SDK REAL-TIME Developed a chat SDK for businesses using React Native and React/Redux, powered by XMPP and a Java backend for real-time messaging. REACTREDUXXMPPJAVA Zero DOWNTIME Built CI/CD automations ensuring continuous deployment with no service interruption. PostgreSQL with Hasura as the GraphQL layer. POSTGRESQLHASURACI/CD
CONSUMER HEALTHCARE Escale SR SOFTWARE ENGINEER JAN 2018 — SEP 2018 SÃO PAULO	Healthcare Platform REACT Built the product foundation with React, Redux, and a Node.js/TypeScript backend for healthcare services sold via Google Ads, including lead capture, eligibility checks, and provider matching. REACTREDUXNODE.JSTYPESCRIPT Mentorship TECH TALKS Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture. MySQL, PostgreSQL, and MongoDB depending on service responsibility. MICROSERVICESCIRCLECIMONGODB
STUDENT LOANS FINTECH PraValer PL SOFTWARE ENGINEER JUN 2017 — JAN 2018 SÃO PAULO	Loan Origination NODE Migrated legacy PHP loan origination platform to a Node.js/Express backend for a student loans product with governmental funding, handling loan request lifecycle, eligibility, and disbursement flows. NODE.JSEXPRESSLOANSPHP Database OPTIMIZATION Restructured PostgreSQL transactional tables with improved schemas and indexes for loan ledger entries. MongoDB for loan requests and logs. Frontend with React and Redux. POSTGRESQLMONGODBREACTREDUX

CORE SKILLS PAYMENTS & FINTECH Plaid, Treasury Prime, ACH, KYC/EDD, FinCEN, Double-entry ledgers, Idempotency, Webhooks BACKEND TypeScript, NestJS, Node.js, Prisma, PostgreSQL, ts-rest, Zod, GraphQL RELIABILITY BullMQ, Outbox pattern, Event ordering, Redis, Datadog FRONTEND React, Next.js, React Native, TanStack Query INFRASTRUCTURE AWS, GCP, Docker, CI/CD, Nx Monorepo LANGUAGES PORTUGUESE Native ENGLISH C1/C2	EDUCATION Anhanguera BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY · JAN 2016 — JUL 2018
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